

can be included in the main code base. If the designers use a single family of controllers, these BUC and POST can be re-used with little changes, and this mandates that the development of BUC/POST should be done with care.

Design verification and validation testing

The second test is for the purpose of design verification and validation (V&V). This V&V test normally is a customised version of the system software, while BUC/POST focus only on the correct functioning.

V&V test software will be a special customised version of system software. While system software is designed to maximise the performance meeting the design needs, V&V software ensures the system and hardware is fully tested for its functionality and weeds out the faults till all known faults are rectified.

V&V software is structured based on the product's documented requirements. For domains like medical devices and avionics documenting, the test case against every known explicit and implicit requirement is a must.

This document is typically known as requirement trace matrix (RTM). Basically, this V&V software needs to test every requirement for its normal working and under fault condition.

Importance of testing during design phase

Testing during the design phase is important to ensure that the design is:

- Correct to the specifications and every specification is tested and proven
- Timing of the circuit is as per the design
- Components operate within the specified limits
- Validates the bill of material so that the costing is frozen
- PCBs and the systems can be moved to manufacturing after the clean-up of any corrections that need to be done.

We need to remember that BUC and POST alone will not test the system fully and we have to have the main software and simulated sources (for the sensors and input signals) to validate the whole system. While the design phase tests check the design, these cannot be used for manufacturing as it is, as the need for manufacturing tests

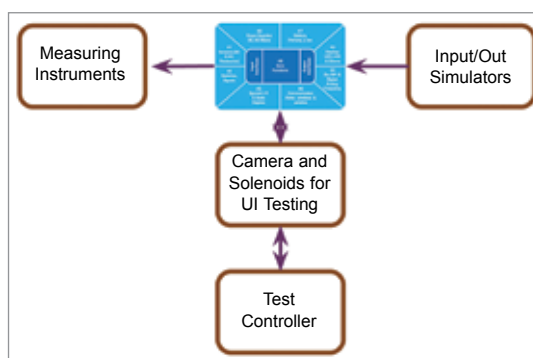


Fig. 5: A typical embedded system test philosophy based on four critical blocks

are different. Some of the manufacturing test needs are:

- Every component needs to be tested individually before the assembled system is tested (this is not the case in the design phase test where the full product is tested).
- Test time has to be very short as with longer testing time product cost will go up due to increased manufacturing costs.
- Manpower available will not have any design knowledge and they are trained to use the tester and just log whether the system has passed the test or not. This means the tests have to be simple with a Go-No Go type of interface.
- Manufacturing tests have to log the test status data automatically without manual intervention so that the tests are efficient and the data is used to control the process.

Before delving into the details of manufacturing tests, let us understand the primary difference between the design phase tests and manufacturing tests.

Design phases are:

- Detailed and exhaustive to ensure the product in its design phase is tested fully for its functionality, performance, and reliability.
- How long the testing takes is not an issue, but the key is that the coverage of the tests should be hundred percent.

• As long as product designers retain the processor/controller from the same family, the POST/BUC tests once developed need only minor changes for the new design. However, the needs for the manufacturing tests are completely different and call for new test jigs.

Manufacturing tests

The primary requirement for testing of products in the manufacturing phase is to ensure the following:

- Every product is tested so that any failure due to component failure and process defects during assembly are detected and weeded out.
- Test the product in the shortest time to improve the manufacturing productivity.
- Ensure every product tested performs consistently

and repeatedly without degradation in performance.

- Ensure every product is calibrated so that it meets the accuracy needed.
- Finally, ensure the cost of testing is kept low to keep the product cost low. Typically, the contribution of testing in product manufacturing is roughly 12-15 percent. Every percentage saved will be a big saving.

Remember that unlike the design phase testing, which is executed one or two times, every product that is produced in the manufacturing line is tested. This means the testing cost in the manufacturing line is a recurring expense and has to be kept minimal. This also means that testability has to be built into the design so that product can be tested fully in the quickest possible time.

Above all, note that in the manufacturing tests the testers may need changes every time the product design is changed. It is even more important that the design has to be stable so that cost incurred due to tester changes are minimised.

A typical embedded system test philosophy is based on four critical blocks as shown in Fig. 5. A typical manufacturing tester has:

1. Test controller which could be a custom-made processor board or a platform like LabView from National Instruments (NI). NI offers a complete package with both hardware and software to create a tester in a modular fashion.

2. Set of voltage, current, and signal sources which generate the signals needed to feed the embedded systems and can be controlled by the test controller.